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Design of Unknown Input Observers and Robust Fault Detection Filters

Adaptive Learning, Estimation and Supervision of Dynamical Systems

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1. Introduction: The Need for Robustness

- **Context:** Model-based Fault Detection and Isolation (FDI) uses residuals to identify system failures.
- **The Core Problem:** Residuals are sensitive not only to faults but also to uncertainties (disturbances, noise, modeling errors).
- **Consequence:** In real-world systems (e.g., aerospace), these uncertainties lead to false alarms and incorrect diagnosis (*mis-isolation*).
- **Proposed Solution:** A strategy that combines **Unknown Input Observers (UIO)** for robustness and **Beard Fault Detection Filters (BFDF)** for isolation.

2. Problem Setting: The Uncertain System

We consider a class of linear systems where uncertainties are modeled as *additive unknown disturbances*:

$$\begin{cases} \dot{\boldsymbol{x}}(t) = \mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\boldsymbol{u}(t) + \mathbf{E}\boldsymbol{d}(t) + \boldsymbol{f}_a(t) \\ \boldsymbol{y}(t) = \mathbf{C}\boldsymbol{x}(t) + \boldsymbol{f}_s(t) \end{cases} \tag{1}$$

Where:

- $\boldsymbol{x} \in \mathbb{R}^n$: State vector; $\boldsymbol{y} \in \mathbb{R}^m$: Output vector; $\boldsymbol{u} \in \mathbb{R}^r$: Known input.
- $\boldsymbol{d} \in \mathbb{R}^q$: Unknown Input vector (disturbances/modeling errors).
- $\mathbf{E}$: Disturbance distribution matrix (assumed full column rank).
- $\boldsymbol{f}_a, \boldsymbol{f}_s$: Actuator and sensor faults.

2. Problem Setting: Modeling Uncertainties

How do we define $\mathbf{E}\boldsymbol{d}(t)$?

Case 1: Parameter Perturbations

If the system matrices have errors $\Delta\mathbf{A}$ and $\Delta\mathbf{B}$:

$$\dot{\boldsymbol{x}} = (\mathbf{A} + \Delta\mathbf{A})\boldsymbol{x} + (\mathbf{B} + \Delta\mathbf{B})\boldsymbol{u} \tag{2}$$

We can approximate the errors as disturbances:

$$\mathbf{E}\boldsymbol{d}(t) \approx \Delta\mathbf{A}\boldsymbol{x} + \Delta\mathbf{B}\boldsymbol{u} \approx \sum a_i \mathbf{A}_i \boldsymbol{x} + \sum b_i \mathbf{B}_i \boldsymbol{u} \tag{3}$$

The matrix $\mathbf{E}$ is constructed using the known structures of $\mathbf{A}_i$ and $\mathbf{B}_i$.

2. Problem Setting: Modeling Uncertainties

Case 2: Non-linearities and Linearization Errors

A non-linear system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$ is linearized around $(\boldsymbol{x}_0, \boldsymbol{u}_0)$.

- Standard Taylor expansion: $f(\boldsymbol{x}, \boldsymbol{u}) \approx \mathbf{A}\Delta\boldsymbol{x} + \mathbf{B}\Delta\boldsymbol{u} +$ Higher Order Terms.
- The Higher Order Terms are often the cause of mis-isolation.
- Strategy: Treat these terms as an unknown input $\boldsymbol{d}(t)$ with a distribution $\mathbf{E}$ estimated from data.

3. The Solution: Structure of the UIO

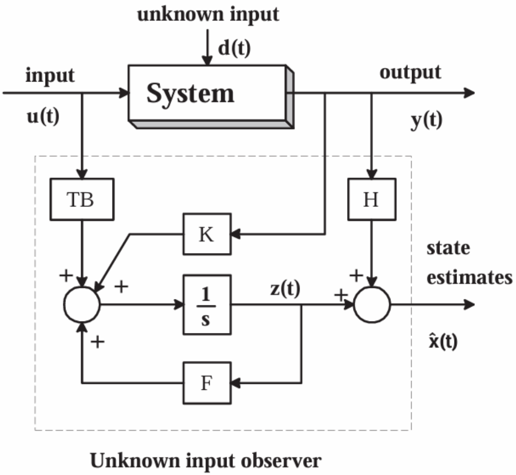
Definition: An observer is defined as an **Unknown Input Observer (UIO)** if its state estimation error vector $\boldsymbol{e}(t)$ approaches zero asymptotically regardless of the presence of the unknown input (disturbance) in the system.

The structure of a full-order observer is:

$$\begin{cases} \dot{\boldsymbol{z}}(t) = \mathbf{F}\boldsymbol{z}(t) + \mathbf{T}\mathbf{B}\boldsymbol{u}(t) + \mathbf{K}\boldsymbol{y}(t) \\ \hat{\boldsymbol{x}}(t) = \boldsymbol{z}(t) + \mathbf{H}\boldsymbol{y}(t) \end{cases} \tag{4}$$

Where $\boldsymbol{z}$ is the observer state and $\hat{\boldsymbol{x}}$ is the estimate state. The goal is to find matrices $\mathbf{F}, \mathbf{T}, \mathbf{K}, \mathbf{H}$ such that the estimation error $\boldsymbol{e} = \boldsymbol{x} - \hat{\boldsymbol{x}}$ is independent of $\boldsymbol{d}$.

3. The Solution: Structure of the UIO



3. The Solution: Achieving Disturbance Decoupling

By substituting the observer equations into the error $\boldsymbol{e} = \boldsymbol{x} - \hat{\boldsymbol{x}}$, we want to make the error dynamics **autonomous** (independent of $\boldsymbol{d}$):

$$\dot{\boldsymbol{e}}(t) = \mathbf{F}\boldsymbol{e}(t) \tag{5}$$

From Theorem 1: To eliminate $\boldsymbol{d}(t)$, we must satisfy $(\mathbf{HC} - \mathbf{I})\mathbf{E} = 0$. This is solvable *if and only if*:

$$\text{rank}(\mathbf{CE}) = \text{rank}(\mathbf{E}) \tag{6}$$

The Intuition:

The number of independent measurements (sensors) must be $\geq$ the number of independent unknown disturbances. We must "see" the disturbances to decouple them.

Once $\boldsymbol{d}(t)$ is decoupled, the error dynamics matrix becomes:

$$\mathbf{F} = \mathbf{A}_1 - \mathbf{K}_1\mathbf{C} \tag{7}$$

Where $\mathbf{A}_1$ is fixed by the decoupling step, but $\mathbf{K}_1$ is a gain matrix we must design to stabilize $\mathbf{F}$.

The core idea of the paper:

- $\mathbf{K}_1$ is *not unique*. We have remaining **design freedom**.
- **Robustness (UIO):** Achieved by the decoupling step.
- **Isolation (BFDF):** We exploit the leftover freedom in $\mathbf{K}_1$ to assign a specific directional signature to the residual $\boldsymbol{r}(t)$ for each possible fault.

3. The Solution: BFDF Principles

A standard observer only tells us *if* a fault occurred. A **Beard Fault Detection Filter** (BFDF) tells us *where* it occurred.

In a system without disturbances, a Beard Fault Detection Filter (BFDF) error dynamics is:

$$\begin{cases} \dot{\boldsymbol{e}} = (\mathbf{A} - \mathbf{KC})\boldsymbol{e} + \boldsymbol{l}_i \eta_i(t) \\ \boldsymbol{r} = \mathbf{C}\boldsymbol{e} \end{cases} \tag{8}$$

Where $\boldsymbol{l}_i$ is the specific fault event direction and $\eta_i(t)$ is its time-evolution.

To isolate the fault, the residual $\boldsymbol{r}(t)$ must maintain a **fixed direction** in the output space. The controllable subspace of the fault must have rank 1:

$$\text{rank}[\boldsymbol{l}_i, (\mathbf{A} - \mathbf{KC})\boldsymbol{l}_i, \ldots, (\mathbf{A} - \mathbf{KC})^{n-1}\boldsymbol{l}_i] = 1 \tag{9}$$

The Intuition:

The fault signature becomes a unique, constant vector (a "fingerprint"), regardless of how the fault magnitude $\eta_i(t)$ changes over time.

By merging the UIO and the BFDF, the error system under a fault becomes:

$$\begin{cases} \dot{\boldsymbol{e}}(t) = (\mathbf{A}_1 - \mathbf{K}_1\mathbf{C})\boldsymbol{e}(t) + \mathbf{T}\boldsymbol{l}_i\eta_i(t) \\ \boldsymbol{r}(t) = \mathbf{C}\boldsymbol{e}(t) \end{cases} \tag{10}$$

This equation represents the ultimate goal of the paper:

- **Robustness via UIO:** The unknown disturbance $\boldsymbol{d}(t)$ is completely decoupled and eliminated from the equation.
- **Isolation via BFDF:** The matrix $\mathbf{K}_1$ is specifically designed to force the residual $\boldsymbol{r}(t)$ to be parallel to the new signature direction $\mathbf{C}\mathbf{T}\boldsymbol{l}_i$.

3. The Solution: Design Procedure Step-by-Step

To build the robust filter, we follow a systematic 5-step procedure:

1. **Check Solvability:** Verify that $\text{rank}(\mathbf{CE}) = \text{rank}(\mathbf{E})$.
2. **Compute Decoupling Matrices:** Calculate $\mathbf{H}^* = \mathbf{E}[(\mathbf{CE})^T\mathbf{CE}]^{-1}(\mathbf{CE})^T$ and $\mathbf{T} = \mathbf{I} - \mathbf{H}^*\mathbf{C}$.
3. **Project System Dynamics:** Compute the new system matrix $\mathbf{A}_1 = \mathbf{A} - \mathbf{H}^*\mathbf{CA}$.
4. **Design Directional Gain:** Use standard BFDF theory to find $\mathbf{K}_1$ such that $\mathbf{C}e$ is uni-directional for each fault $\mathbf{T}\boldsymbol{l}_i$ and $(\mathbf{A}_1 - \mathbf{K}_1\mathbf{C})$ is stable.
5. **Final Observer Gain:** Calculate the total gain $\mathbf{K} = \mathbf{K}_1 + \mathbf{FH}^*$.

3. The Solution: Quantifying Actuator Faults

How do we mathematically measure if the residual $\boldsymbol{r}(t)$ is pointing in the right direction?

For an **actuator fault**, the signature direction is a 1D vector $\boldsymbol{v}_i = \mathbf{CT}\boldsymbol{l}_i$. We use the *Correlation Parameter*:

$$CORR_i(t) = \frac{|\boldsymbol{v}_i^T \boldsymbol{r}(t)|}{\|\boldsymbol{v}_i\|_2 \, \|\boldsymbol{r}(t)\|_2} \tag{11}$$

This is simply the absolute cosine of the angle between the residual and the theoretical signature direction.

- $CORR_i \approx 1$: Perfect alignment $\implies$ **High confidence in fault** i.
- $CORR_i \approx 0$: Orthogonal $\implies$ **Fault** i **is not occurring**.

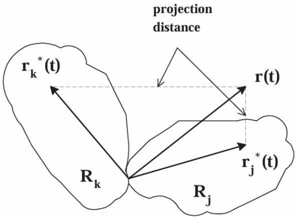
3. The Solution: Isolating Sensor Faults

Sensor faults are more complex, their signature doesn't lie on a single line, but spans a plane/subspace R_j:

$$R_j = \text{Span}\{\boldsymbol{I}_j, \mathbf{C}\boldsymbol{k}_{1j}, \mathbf{C}\boldsymbol{h}_j\} \tag{12}$$

To check if the residual belongs to this subspace, we compute the **Normalized Projection Distance (NPD)**:

$$NPD_j = \frac{\left\| \boldsymbol{r}(t) - \boldsymbol{r}_j^*(t) \right\|_2}{\|\boldsymbol{r}(t)\|_2} \tag{13}$$



Where $\boldsymbol{r}_j^*(t)$ is the projection of $\boldsymbol{r}(t)$ onto R_j.

Decision Rule: The smallest NPD_j identifies the faulty sensor j.

4. The Paper's Case Study: Jet Engine Model

The authors apply their theory to a non-linear jet engine control system.

State Variables ($\boldsymbol{x} \in \mathbb{R}^3$):

- x_1 (n_L): Low pressure rotor speed
- x_2 (n_H): High pressure rotor speed
- x_3 (W_f): Main burner fuel flow

Control Input ($u \in \mathbb{R}$):

- u (W_{fe}): Fuel flow command

Measurements ($\boldsymbol{y} \in \mathbb{R}^6$):

$\boldsymbol{y} = [x_1,\ x_2,\ x_3,\ p_2,\ p_4,\ t_4]^T$

- p_2: High pressure compressor discharge pressure
- p_4: Turbine discharge pressure
- t_4: Turbine exit temperature

The system is linearized around an equilibrium point ($n_L = 450$ rpm) to extract the baseline matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$. Both filters use this linear approximation as their fundamental "knowledge" of the plant.

A linear model is imperfect. The authors represent the linearization errors (the non-linearities) as an additive unknown disturbance:

$$\dot{\boldsymbol{x}} = \mathbf{A}\boldsymbol{x} + \mathbf{B}\boldsymbol{u} + \mathbf{E}\boldsymbol{d}(\boldsymbol{x}) \tag{14}$$

- $\boldsymbol{d}(\boldsymbol{x})$ contains the second-order Taylor expansion terms: $[x_1^2, x_2^2, x_3^2, x_1x_2, \dots]^T$.
- **Matrix Identification:** Using steady-state data from a highly complex non-linear simulator, the authors used Least-Squares to estimate the distribution matrix $\mathbf{E}^*$.
- **Decoupling Capacity:** They extracted a full column rank matrix $\mathbf{E}_1$ (rank 2). Since we have 6 sensors, rank$(\mathbf{CE}_1) = 2$, leaving enough design freedom for fault isolation.

The authors then used $\mathbf{E}_1$ to compute the robust filter matrices $\mathbf{H}, \mathbf{T}, \mathbf{K}$.

5. Our Experimental Replication: The Surrogate Model

Challenge: The original paper tested the filters on a proprietary, black-box non-linear NASA simulator. To replicate the results in MATLAB, we build a **Surrogate Non-Linear Model**.

Simulation Approach: we programmed the plant dynamics using the nominal matrices *plus* the quadratic Taylor terms mapped by the authors' $\mathbf{E}^*$ matrix.

$$\dot{\boldsymbol{x}}_{sim} = \mathbf{A}\boldsymbol{x} + \mathbf{B}\boldsymbol{u} + (\text{Distortion Factor}) \times \mathbf{E}^*\boldsymbol{d}(\boldsymbol{x}) \tag{15}$$

The "Stress Test" (Distortion Factor = 20): To emulate the severe non-linearities of the real simulator and force the Standard BFDF to fail (as it did in the paper), we amplified the error term. This proves the robustness of the UIO under harsh modeling mismatches.

5. Our Experimental Replication: Test Scenario

We ran two filters in parallel against the Surrogate Non-Linear Plant:

1. **Standard BFDF:** Designed using standard pole-placement on $\mathbf{A}, \mathbf{C}$. It has *no knowledge* of the non-linearities.
2. **Robust UIO:** Implemented using the exact $\mathbf{H}, \mathbf{T}, \mathbf{K}$ matrices computed by the authors. It mathematically decouples the non-linearities.

Simulation Parameters:

- **Injected Fault:** $+2\%$ magnitude offset on **Sensor 1**.
- **Timing:** System starts healthy, fault occurs at $t = 3.0$ seconds.
- **Evaluation:** Calculate the dynamic NPD_j for both filters to see which one correctly isolates Sensor 1.

6. Results: Reproducing Table 1 and Table 2

Standard BFDF (Table 1)

Faulty Sensor	No.1	No.2	No.3
NPD_1	0.81846	0.76787	0.85425
NPD_2	**0.61743**	**0.33689**	0.66355
NPD_3	0.98809	0.98411	**0.43531**

Note: Fault on S1 is mis-isolated as S2.

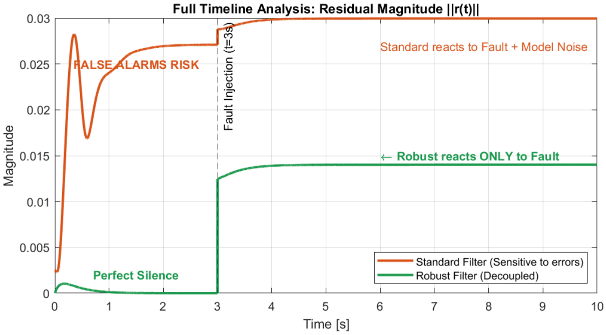
Robust UIO (Table 2)

Faulty Sensor	No.1	No.2	No.3
NPD_1	**0.01340**	0.41593	0.21472
NPD_2	0.48555	**0.02352**	0.20938
NPD_3	0.85629	0.70354	**0.00113**

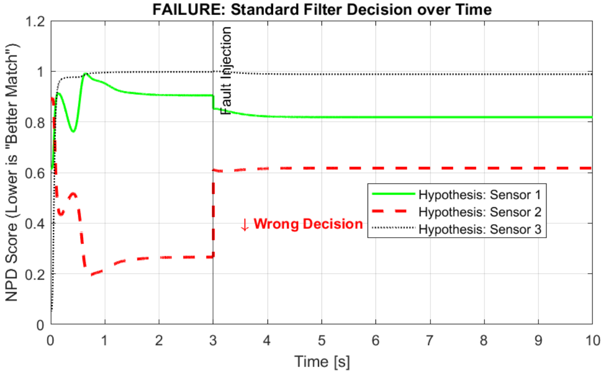
Note: All faults are correctly isolated.

Decoupling in Action

- This graph shows why the Robust Filter is superior before the fault even happens.
- The Orange line shows the Standard Filter reacting to model noise (false alarms).
- The Green line remains flat at zero, proving perfect disturbance decoupling.



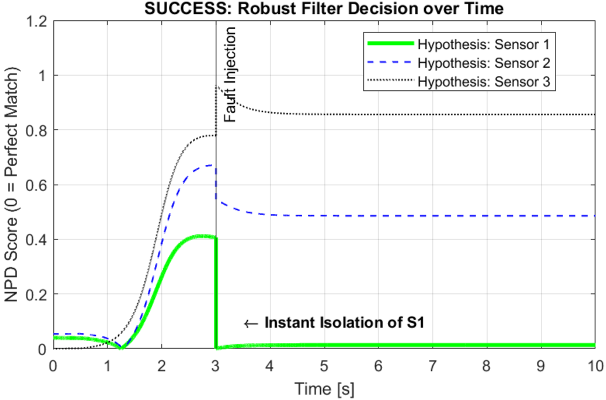
6. Results: Standard Filter Chaos



Failure Analysis

- Here we see the decision process over time for the Standard Filter.
- Notice how the Red line (Hypothesis S2) drops below the Green line (Hypothesis S1).
- The filter is confused by the non-linearities.

6. Results: Robust Filter Precision



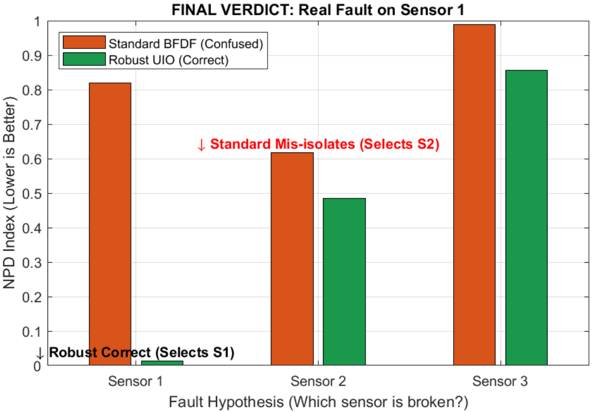
Success Analysis

- In contrast, look at the Robust Filter.
- The Green line (Hypothesis S1) drops instantly to zero at the moment of the fault.
- The decision is immediate and unambiguous.

6. Results: The Final Verdict

Real fault on Sensor 1

- The Standard Filter (orange) misidentifies the fault on Sensor 1 as a fault on Sensor 2 (lowest bar is wrong).
- The Robust Filter (green) correctly identifies Sensor 1 (lowest bar is correct).



7. Conclusion: Project Summary & Key Takeaways

- **Objective Achieved:** Successfully reproduced the Robust Fault Detection Filter by mathematically merging UIO with BFDF.
- **The Danger of Non-Linearities:** We demonstrated that in non-linear environments (like the Jet Engine), standard filters are blinded by modeling errors, leading to dangerous mis-isolation.
- **The Power of Decoupling:** By identifying the disturbance matrix $\mathbf{E}$ and decoupling it, the Robust Filter restores **perfect isolation accuracy**, with identical online computational cost.

References

- Chen, J., Patton, R. J., Zhang, H. Y. "Design of Unknown Input Observers and Robust Fault Detection Filters".